DIVINE GRACE COLLEGE OF ARTS, SCIENCE AND TECHNOLOGY (DGCAST) – GAROUA

GEOGRAPHY REVISION SHEET – FORM FOUR

First Sequence Evaluation, School Year 2026-2027 • Teacher: Mr KAMDEM Emmanuel

Lessons revised: Part 1: physical geography (Form 3 syllabus, lessons 1 to 25); Part 2: soils, ecosystems, biomes, Cameroon (Mr Tabit)

PART 1: PHYSICAL GEOGRAPHY (Mr KAMDEM) – FORM 3 SYLLABUS, LESSONS 1 TO 25 – FOR THE 50-MINUTE SESSION

1. Time and the movements of the Earth: how many degrees make one hour? Give the effects of rotation and revolution.

Answer: 360° in 24 hours: **15° = 1 hour, 1° = 4 minutes**; places in the east are ahead. Example: 12:00 at Garoua (13° E) is 13:20 at 33° E (20° × 4 = 80 min). **Rotation** (west to east, 24 h): day and night, time differences, deflection of winds and tides. **Revolution** (365¼ days): seasons, varying day length, leap year.

A. STRUCTURE OF THE EARTH, ROCKS AND PLATES (LESSONS 7 TO 10)

2. Name the layers of the Earth and the three types of rocks.

Answer: Crust (solid, thin), **mantle** (very hot rock), **outer core** (liquid iron and nickel), **inner core** (solid iron and nickel). Rocks: **igneous** (magma cools: granite, basalt), **sedimentary** (layers of sediments: sandstone, limestone), **metamorphic** (changed by heat and pressure: marble, slate).

3. Continental drift and plate tectonics: author, evidence and plate margins.

Answer: Wegener (1912): one super-continent, **Pangaea**. Evidence: jigsaw fit of the coasts, same fossils, same rocks and mountains, ice marks in hot lands. Plate margins: **constructive** (plates move apart: ridges, rift valleys), **destructive** (plates move together: fold mountains, trenches, volcanoes), **conservative** (plates slide past: faults).

B. VOLCANOES AND EARTHQUAKES (LESSONS 11 TO 13)

4. Volcanoes: types, parts, benefits and hazards.

Answer: Active (Mount Cameroon), dormant, extinct (Manengouba). Parts: magma chamber, vent, cone, crater. Benefits: fertile soils, geothermal energy, tourism. Hazards: lava, ash, poisonous gases (Lake Nyos, 1986), lahars. Measures: monitoring, evacuation, land-use planning.

5. Earthquakes: focus, epicentre, causes, effects and measures.

Answer: The **focus** is where the earthquake starts; the **epicentre** is the point on the surface above it. Cause: mainly plate movements along faults. Measured with a seismograph (Richter scale). Effects: loss of life, destroyed buildings, fires, tsunamis. Measures: strong buildings, warning, drills.

C. WEATHERING (LESSONS 14 TO 17)

6. Define weathering. Distinguish physical from chemical weathering and name their processes.

Answer: Weathering is the breaking down of rocks in their place. **Physical**: rocks break without change of minerals: thermal shattering, **frost shattering**, wetting and drying, pressure release. **Chemical**: minerals are changed by water and air: solution, carbonation, hydrolysis, oxidation, hydration.

D. RIVERS (LESSONS 18 TO 22)

7. Describe the three courses of a river and their landforms.

Answer: Upper: steep, vertical erosion; V-shaped valley, interlocking spurs, waterfall, gorge. **Middle**: gentle slope, lateral erosion; **meanders** (river cliff on the outside, slip-off slope on the inside). **Lower**: almost flat, deposition; floodplain, levees, ox-bow lake, delta. Erosion: hydraulic action, abrasion, attrition, solution.

8. Man and the floodplain: importance, problems and solutions.

Answer: Importance: fertile alluvium, water, fishing, energy (Lagdo Dam), transport. Problems: floods, diseases, displacement. Solutions: dykes, dams, warnings, building on higher ground.

E. WAVE ACTION (LESSONS 23 TO 25)

9. Waves: types, erosional and depositional features.

Answer: Constructive waves (gentle, strong swash) build beaches; **destructive** waves (high, strong backwash) erode. Erosion: capes and bays, cliff and wave-cut platform, then cave → arch → stack → stump. Deposition: **longshore drift** moves sand along the coast and forms a beach, a **spit** (strip of sand joined to the land at one end), a bar and a tombolo.

Quick quiz	Answer
15° of longitude =	**1 hour**
Liquid layer of the Earth	**Outer core**
Plates move apart at …	**Constructive margins**
Strip of sand joined to the land at one end	**Spit**

PART 2: FORM 4 SYLLABUS (Mr TABIT) – FOLLOWS, IF TIME ALLOWS

A. THE SOIL: MEANING, COMPONENTS AND PROFILE

1. Define soil. What is it made of?

Answer: Soil is the loose top layer of the Earth's surface in which plants grow. It is formed by physical, chemical and biological processes and is made of **mineral particles** (about 45 %), **organic matter** or humus (about 5 %), and **water and air** (about half of the soil).

2. Name the horizons of a simple soil profile.

Answer: Topsoil (A): dark, rich in humus, most roots; **subsoil (B)**: little humus, minerals washed down from the topsoil; **regolith (C)**: weathered and broken rock; **bedrock (R)**: solid rock at the bottom. A deep soil rich in humus and minerals is fertile.

B. SOIL EROSION AND SOIL CONSERVATION

3. Define soil erosion and accelerated erosion. Name the forms of erosion.

Answer: Soil erosion is the removal of the topsoil by water, wind or people. **Accelerated erosion** is erosion made faster by human activities, while normal (geologic) erosion is slow and natural. Forms: sheet, rill, gully and wind erosion.

4. Give the causes and the consequences of soil erosion.

Answer: Physical causes: heavy rain, steep slopes, strong winds. Human causes: deforestation, overgrazing, bush fires, ploughing up and down slopes, over-cultivation. Consequences: the soil loses its fertility and harvests fall; gullies destroy land and roads; rivers and dams fill with mud.

5. Give methods of soil conservation.

Answer: Mechanical: **terracing** (flat steps cut into a slope, as on the Mandara Mountains), contour ploughing and ridges, stone lines and check dams. Biological: planting trees and **windbreaks**, cover crops, mulching, crop rotation and manure. Other: controlled grazing and stopping bush fires. **Strip cultivation** (strips of crops across the slope) also protects the soil.

C. THE ECOSYSTEM

6. Define an ecosystem and give its components.

Answer: An **ecosystem** is a natural area where living and non-living components interact. Components: abiotic (sunlight, water, air, soil, temperature); **producers** (green plants that make their own food from solar energy); consumers (animals); decomposers (fungi and bacteria).

7. How does energy flow in an ecosystem?

Answer: Energy comes from the Sun and is captured by producers, then passes to herbivores and carnivores. Only about **10 %** of the energy passes from one trophic level to the next; the rest is lost as heat. Example of a food chain: grass → zebra → lion; many chains linked make a food web.

D. CLIMATE DATA, SAVANNA AND HOT DESERT BIOMES

8. How do you work from a climate table?

Answer: Annual range of temperature = highest − lowest monthly temperature. Total rainfall = sum of the 12 monthly figures. The wettest month has the highest rainfall. Then compare with the biomes: rainforest (rain all year, range 2–3 °C), savanna (one wet and one dry season, 750–1,500 mm), hot desert (below 250 mm).

9. Station W (24–29 °C; one rainy season May to October; about 1,070 mm; wettest month August): name its biome with two reasons.

Answer: Tropical wet and dry (savanna): hot all year with a small range (5 °C), one long rainy season and a long dry season, rainfall of about 1,000 mm.

10. Describe the savanna (location, vegetation, soils) and how its vegetation adapts to the dry season.

Answer: Between about 5° and 15° N and S, including northern Cameroon; tall grass with scattered trees; red ferruginous soils. Adaptations: the **baobab** stores water in its trunk; many trees shed their leaves; deep roots and thick bark against fire; grasses dry up and grow again with the first rains.

11. Describe the hot desert biome. Name its depositional features and ways to solve water scarcity.

Answer: Less than 250 mm of rain, very hot days and cold nights, sandy and stony soils, xerophytes (cacti, acacia). Wind deposition forms dunes: **barchans** (crescent-shaped) and seif; inselbergs, rock pedestals and yardangs are erosional features. Water scarcity: build **water reservoirs**, dig wells and boreholes, irrigate oasis farms.

E. CAMEROON: SITUATION AND RELIEF

12. Where is Cameroon located?

Answer: Between about **2° N and 13° N** and **8° E and 16° E**; in Central Africa, on the Gulf of Guinea. It covers about 475,000 km², is triangular in shape and has about 400 km of Atlantic coast.

13. Name the relief units of Cameroon.

Answer: Highlands: Western Highlands (Mount Cameroon, 4,095 m, the highest point), Adamawa Plateau (1,000–1,500 m), Mandara Mountains, South Cameroon Plateau. Lowlands: Coastal lowlands (Douala, Kribi), Benue lowland (Garoua), Chad Plain.

14. Competence: a widow's farm on a steep hill slope loses soil to wind and rain and her yields fall. State the problem, two causes and two measures.

Answer: Problem: soil erosion by rain and wind, so yields fall. **Causes**: farming a steep slope without protection; clearing the plant cover; ploughing up and down the slope. **Measures**: terracing and contour ploughing; planting trees, windbreaks and cover crops; mulching.